

Selected topics

Random numbers



Random numbers (1/4)

Float number in $[0;1]$:

```
rand()
```

```
julia> rand()
```

Vector of n floats in $[0;1]$:

```
rand( $n$ )
```

```
julia> rand(3)
```

Matrix of $n_1 \times n_2$ floats in $[0;1]$:

```
rand( $n_1, n_2$ )
```

```
julia> rand(3,2)
```

Random numbers (1/4)

Float number in $[0;1]$:

```
rand()
```

```
julia> rand()
```

Vector of n floats in $[0;1]$:

```
rand( $n$ )
```

```
julia> rand(3)
```

Matrix of $n_1 \times n_2$ floats in $[0;1]$:

```
rand( $n_1, n_2$ )
```

```
julia> rand(3,2)
```

Random numbers (1/4)

Float number in $[0;1]$:

```
rand()
```

```
julia> rand()
```

Vector of n floats in $[0;1]$:

```
rand( $n$ )
```

```
julia> rand(3)
```

Matrix of $n_1 \times n_2$ floats in $[0;1]$:

```
rand( $n_1, n_2$ )
```

```
julia> rand(3,2)
```

Random numbers (2/4)

Integer number in $[a;b]$:

```
rand(a:b)
```

```
julia> rand(1:6)
```

Vector of n integers in $[a;b]$:

```
rand(a:b,n)
```

```
julia> rand(1:6,2)
```

Matrix of $n_1 \times n_2$ integers in $[a;b]$:

```
rand(a:b,n1,n2)
```

```
julia> rand(1:6,3,2)
```

Random numbers (2/4)

Integer number in $[a;b]$:

```
rand(a:b)
```

```
julia> rand(1:6)
```

Vector of n integers in $[a;b]$:

```
rand(a:b,n)
```

```
julia> rand(1:6,2)
```

Matrix of $n_1 \times n_2$ integers in $[a;b]$:

```
rand(a:b,n1,n2)
```

```
julia> rand(1:6,3,2)
```

Random numbers (3/4)

```
julia> using Random
```

A random permutation of length n :

```
randperm( $n$ )
```

```
julia> randperm(5)
```

Return a randomly permuted copy of vector v :

```
shuffle( $v$ )
```

```
julia> shuffle(Vector(1:10))
```

Random numbers (3/4)

```
julia> using Random
```

A random permutation of length n :

```
randperm( $n$ )
```

```
julia> randperm(5)
```

Return a randomly permuted copy of vector v :

```
shuffle( $v$ )
```

```
julia> shuffle(Vector(1:10))
```


Random numbers (4/4)

Reseed the random number generator:

```
Random.seed!(init)
```

```
julia> Random.seed!(1234)
julia> rand()
julia> rand()
julia> Random.seed!(1234)
julia> rand()
julia> rand()
```

→ a reproducible sequence of numbers when a seed is provided.

